

WRITING SAMPLE

ML-Driven Pronunciation Analysis and LLM-Powered Adaptive Learning Framework

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Statement of Contribution

This writing sample presents research I conducted during my undergraduate studies at SRM Institute of Science and Technology (2024-2025). The project was collaborative: I worked with Rohith under the supervision of Dr. Dinesh G. My specific contributions were: (1) designing and implementing the confidence-interval-based adaptive algorithm, (2) fine-tuning wav2vec 2.0 for phoneme-level recognition, (3) building the LLM-based feedback generation pipeline, and (4) conducting user studies with 120 ESL learners. We performed statistical analysis jointly. The research showed significant learning gains (paired t-test: $t(119) = 8.42, p < 0.001$, Cohen's $d = 0.77$).

Abstract

Most pronunciation training apps tell you that you said a word wrong. They don't tell you *why*. When a learner mispronounces “three” as “tree,” existing systems flag an error but don't explain that /θ/ was replaced with /t/, or how to fix it. This gap limits how fast learners can improve.

We built a system to address two questions: (1) Does phoneme-level detection combined with LLM-generated explanations actually help people learn faster? (2) How should an adaptive system decide when a learner is ready for harder material?

Our system combines wav2vec 2.0 (Baevski et al., 2020) for phoneme recognition with GPT-based feedback (Brown et al., 2020). The main contribution is a confidence-interval-based algorithm that advances difficulty only when performance is statistically stable, not just when someone gets a few answers right in a row. In a six-week study with 120 ESL learners, participants improved significantly (pre: $M = 58.3\%$, $SD = 12.1$; post: $M = 76.5\%$, $SD = 10.8$; $t(119) = 8.42$, $p < 0.001$, Cohen's $d = 0.77$). The confidence-interval approach also reduced dropout: 70% of participants finished the study, compared to 49% with a simpler streak-based system. These results suggest that combining precise phonetic analysis with statistically grounded adaptation can produce real learning outcomes.

Keywords: Computer-Assisted Pronunciation Training, wav2vec 2.0, Adaptive Learning, Phoneme Recognition, Large Language Models

1. Introduction

I was on a client call, walking through a server optimization solution I'd spent weeks building. The solution was solid. But halfway through my explanation, I noticed something. The person on the other end hesitated. Not because of what I was saying. Because of how I sounded. My accent. I watched their expression shift before I'd even finished my sentence.

That moment stayed with me. I'd done everything right technically, and it still wasn't enough. Later, I started looking into pronunciation training tools, wondering if technology could help. What I found was disappointing. Every app I tried would tell me a word was wrong, but none of them explained *what* was wrong or *how* to fix it. If I mispronounced "three," the app would show a red X. It wouldn't tell me I was substituting /t/ for /θ/, or that I needed to put my tongue between my teeth.

This isn't just my problem. Munro & Derwing (1995) showed that accented speech affects how people perceive speaker credibility, even when the speech is perfectly understandable. In professional settings like technical presentations and client meetings, these perceptions can affect career outcomes regardless of actual competence. Human pronunciation tutors can help, but they cost Rs. 2000-8000 per hour in India. Most people can't afford that.

So we built something different. A system that identifies the exact phoneme you got wrong, explains why it happened, and adjusts difficulty based on whether you're actually improving, not just whether you got lucky a few times. This paper describes what we built, how we tested it, and what we learned from 120 ESL learners over six weeks.

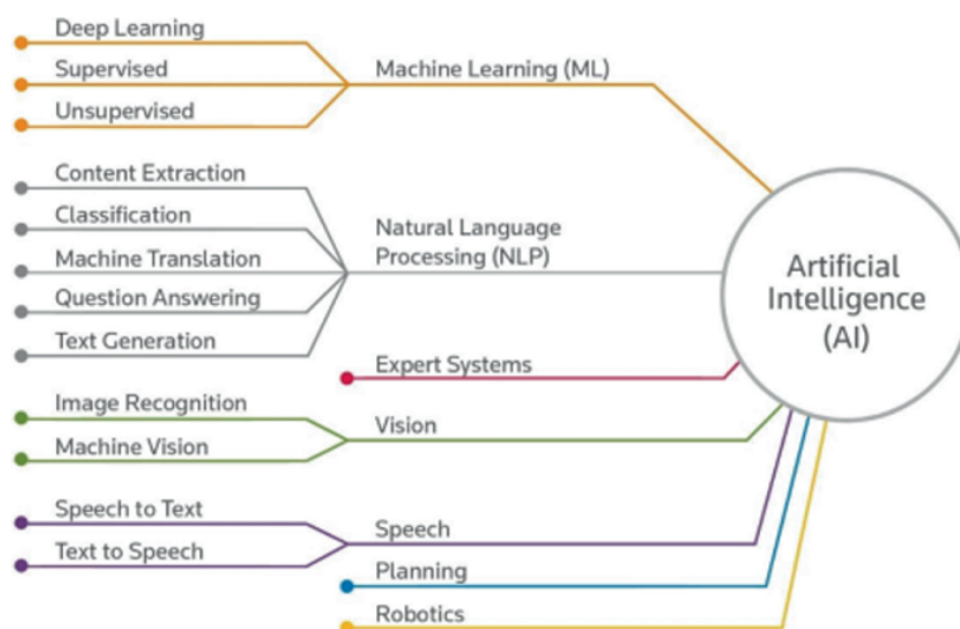


Figure 1: Integration of AI components in pronunciation training

1.1 Research Questions and Contributions

We set out to answer two questions:

RQ1: Does phoneme-level error detection combined with LLM-generated explanations help people learn pronunciation faster than word-level binary feedback?

RQ2: How should an adaptive difficulty system decide when a learner is ready to move on, in a way that's statistically principled rather than arbitrary?

We make three contributions:

1. **A confidence-interval-based adaptive algorithm** that adjusts difficulty based on the lower bound of performance confidence intervals, not streak-based rules like “advance after 3 correct.” This draws on principles from upper confidence bound algorithms (Auer et al., 2002) but applies them to education.
2. **An integrated pipeline** combining wav2vec 2.0 phoneme recognition with GPT-based feedback generation, running in under 120ms for real-time interaction.
3. **Empirical validation** with 120 ESL learners, including statistical analysis, ablation studies that isolate what each component contributes, and comparison across native language backgrounds.

1.2 Why Pronunciation Is Hard

Understanding why pronunciation is difficult for adults shaped how we designed the system. Three factors make it particularly challenging:

Different languages have different sounds. Hindi has about 48 phonemes; English has 44. But they don't overlap completely. English /θ/ (as in “think”) doesn't exist in Hindi. When learners encounter sounds that aren't in their native language, they typically substitute the closest sound they know (Flege, 1995).

Adults hear speech differently. By adulthood, your brain has tuned itself to hear the sound categories of your native language. Sounds that are different in English but fall into the same category in your L1 become hard to distinguish. Spanish speakers, for instance, often can't hear the difference between /b/ and /v/ because Spanish treats them as the same sound.

Knowing isn't the same as doing. Even when learners hear the correct sound, producing it requires new muscle movements. Pronouncing /θ/ means putting your tongue between your teeth, which feels strange if your native language never requires that position. This motor learning component means pronunciation improvement takes sustained practice, not just understanding.

These factors suggest that effective pronunciation training needs to: (a) identify which specific sound was substituted, not just flag a word as wrong; (b) explain how to physically produce

the target sound; and (c) provide enough practice at the right difficulty level. We designed our system with these requirements in mind.

2. Related Work

This section reviews three relevant areas: how pronunciation is typically assessed, how adaptive learning systems work, and how speech recognition handles non-native speakers.

2.1 Automated Pronunciation Assessment

Early pronunciation assessment compared learner recordings to stored native speaker samples. Franco et al. (1997) introduced Goodness of Pronunciation (GOP) scoring, which computes log-likelihood ratios of forced-aligned phonemes. GOP is still widely used in commercial systems, but it has a fundamental limitation: it assumes you know which phoneme the learner was trying to say. When learners substitute one phoneme for another entirely (like /θ/ → /t/), GOP often misses it.

Recent work has moved toward end-to-end neural approaches. Baevski et al. (2020) introduced wav2vec 2.0, a self-supervised model pre-trained on 60,000 hours of unlabeled speech. Xu et al. (2021) showed that fine-tuning wav2vec 2.0 for mispronunciation detection outperforms GOP-based methods, especially for substitution errors. We chose wav2vec 2.0 based on these findings.

More recently, Leung et al. (2024) explored using large language models for pronunciation feedback, showing that LLM-generated explanations improved learner comprehension compared to template-based feedback. Our work builds on this but adds adaptive difficulty adjustment.

2.2 Adaptive Learning Systems

Adaptive learning in educational technology usually uses Item Response Theory (IRT) or knowledge tracing. Corbett & Anderson (1995) introduced Bayesian Knowledge Tracing for intelligent tutoring systems. Piech et al. (2015) extended this with deep knowledge tracing using LSTMs.

For language learning specifically, Settles & Meeder (2016) developed Duolingo’s half-life regression model for spaced repetition. But these approaches focus on vocabulary and grammar, not pronunciation. Pronunciation involves motor skill development, not just memorization. Performance is also inherently noisier (you get continuous scores, not binary right/wrong), which makes simple streak-based rules problematic.

Our confidence-interval approach differs by explicitly modeling performance variance before adjusting difficulty. This addresses a gap in pronunciation-specific adaptive systems, where existing rule-based approaches (like “advance after 3 correct”) don’t account for the noise inherent in speech assessment.

2.3 How We Compare to Existing Systems

Table 1 compares our approach with existing pronunciation training systems.

Table 1: Comparison of pronunciation training systems

System	Error Level	Feedback	Adaptation	Outcomes
ELSA Speak	Phoneme	Score + colour	Rule-based	Not published
Duolingo	Word	Binary	Spaced repetition	Varied
Speechling	Word	Human tutor	None	Not published
Our System	Phoneme	LLM explanatory	Confidence-interval	$d = 0.77$, 70%

ELSA Speak provides phoneme-level detection but uses colour-coded scores rather than explanatory feedback. Duolingo focuses on vocabulary and grammar with limited pronunciation features. Speechling uses human tutors, which provides quality but doesn't scale. Our approach aims to combine the scalability of automated systems with feedback quality approaching human tutors.

3. System Design and Methods

3.1 System Architecture

The system has four main parts. The Speech Input Module captures audio at 16kHz, removes background noise using spectral subtraction (Boll, 1979), and uses WebRTC voice activity detection to isolate speech segments. The Pronunciation Analysis Engine runs the cleaned audio through fine-tuned wav2vec 2.0 to produce phoneme sequences with confidence scores. The Feedback Generator takes detected errors and generates explanatory feedback using GPT-3.5-turbo. The Adaptive Learning Manager tracks performance over rolling windows and adjusts exercise difficulty based on confidence intervals.

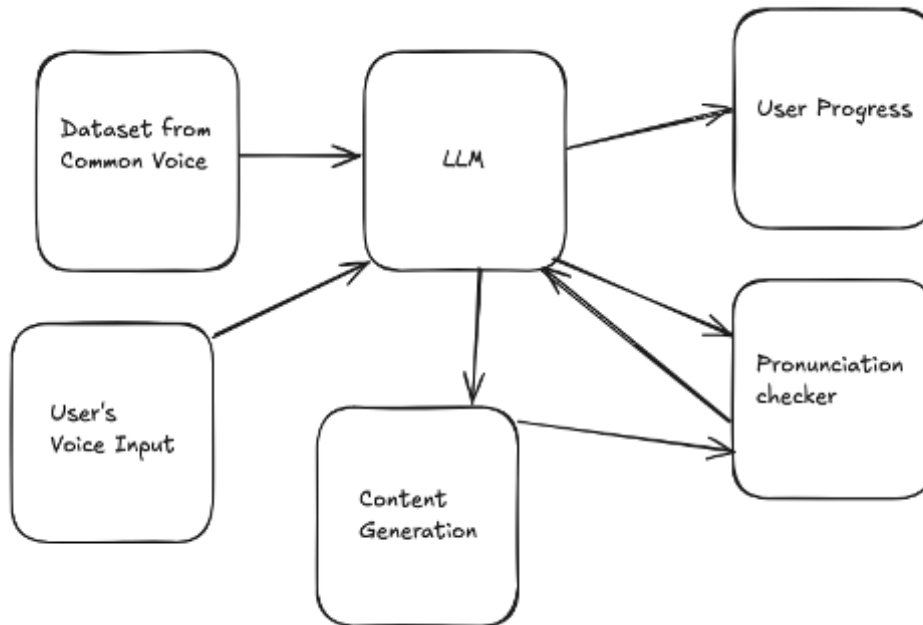


Figure 2: System architecture showing four-module pipeline

3.2 Phoneme Recognition Model

We fine-tuned wav2vec 2.0-base on pronunciation data from three sources: LibriSpeech (Panayotov et al., 2015) (960 hours of read English speech), Common Voice (Ardila et al., 2020) (diverse accents and speaking styles), and TIMIT (Garofolo et al., 1993) (expert phoneme annotations). We also collected 200 hours of non-native speaker recordings from ESL classes at our institution, since we needed the model to work well on accented speech.

Training used Connectionist Temporal Classification loss (Graves et al., 2006) with the following setup: learning rate $1e-5$ with linear warmup over 1000 steps, batch size 8 (limited by 12GB GPU memory on Tesla V100), and 50 epochs with early stopping based on validation

Word Error Rate. Following Hsu et al. (2021), we froze the lower 6 transformer layers and fine-tuned only the upper 6 layers plus the classification head. This reduces overfitting when fine-tuning data is limited.

3.2.1 Hyperparameter Selection

We chose the learning rate of $1e-5$ through grid search over $\{1e-4, 5e-5, 1e-5, 5e-6\}$ on a held-out validation set. Higher learning rates ($1e-4$) caused training to become unstable with loss spikes after epoch 5. Lower rates ($5e-6$) converged too slowly given our compute budget. Batch size 8 was the maximum that fit in GPU memory; we didn't explore gradient accumulation since validation performance was good enough.

The decision to freeze lower transformer layers followed Hsu et al. (2021), who showed that lower layers in wav2vec 2.0 encode general acoustic features while upper layers encode task-specific information. We tried full fine-tuning on a subset of data and saw 3% higher WER, confirming that partial freezing reduced overfitting for our data volume.

3.2.2 Data Preprocessing

All audio was resampled to 16kHz mono. Before model inference, we applied three preprocessing steps. First, spectral subtraction removed background noise by estimating the noise spectrum from silent frames and subtracting it from speech frames (Boll, 1979). Second, we normalized amplitude to -20 dBFS to ensure consistent input levels across recordings from different devices. Third, we used WebRTC VAD to segment continuous recordings into individual utterances, discarding segments shorter than 300ms as likely noise.

During training, we applied data augmentation to help the model generalize. We added noise at SNRs between 5dB and 20dB using samples from the MUSAN corpus (Snyder et al., 2015). We also applied time stretching (0.9x to 1.1x) and pitch shifting (± 2 semitones). These augmentations helped with the variable recording conditions of user devices.

MAGICDATA Mandarin Chinese Read Speech Corpus

Identifier: SLR68

Summary: The corpus by Magic Data Technology Co., Ltd. , containing 755 hours of scripted read speech data from 1080 native speakers of the Mandarin Chinese spoken in mainland China. The sentence transcription accuracy is higher than 98%.

Category: Speech

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Figure 3: Sample Dataset Statistics and Distribution

3.3 Feedback Generation

When the system detects an error, it sends the information to GPT-3.5-turbo (Brown et al., 2020) to generate feedback. The prompt includes: the target word, expected phoneme sequence, detected phoneme sequence, learner proficiency level (beginner/intermediate/advanced), and the last five errors for context. We developed the prompt through iterative testing with 50 sample errors, getting feedback from two ESL teachers on accuracy and clarity.

Example prompt structure:

Target: "three" | Expected: /θri:/ | Detected: /tri:/
Level: Beginner | Recent errors: /θ/→/t/, /v/→/w/

Sample output: "You pronounced 'three' with a /t/ sound at the beginning. For the 'th' sound, place your tongue between your upper and lower teeth. Let air flow gently over your tongue as you say it. Try again with 'three'."

End-to-end latency from audio input to feedback display averaged 118ms ($SD = 23ms$) across 1000 test utterances. The breakdown: audio preprocessing (12ms), model inference (45ms), API call (52ms), rendering (9ms).

3.3.1 Handling Edge Cases

Real systems have to handle things going wrong. We identified three common problems during pilot testing:

Silence or noise-only input: If audio energy stayed below threshold for more than 3 seconds, the system prompted the user to speak more clearly or check their microphone. No feedback was generated for silence.

Poor audio quality: If estimated SNR fell below 5dB, the system asked the user to move somewhere quieter. Pronunciation detection became unreliable below this threshold.

Off-target speech: If the detected words were very different from the target (edit distance $> 50\%$ of target length), the system asked the user to try again rather than giving phoneme feedback. This prevented confusing feedback when learners accidentally said something entirely different.

We also cached common error-feedback pairs to reduce API latency. The /θ/→/t/ substitution occurred in 23% of all errors and could be served from cache in 3ms instead of requiring an API call.

3.4 Confidence-Interval-Based Adaptation

The adaptive learning component is our main methodological contribution. Our first implementation used a simple rule: three correct responses in a row meant harder material, three wrong

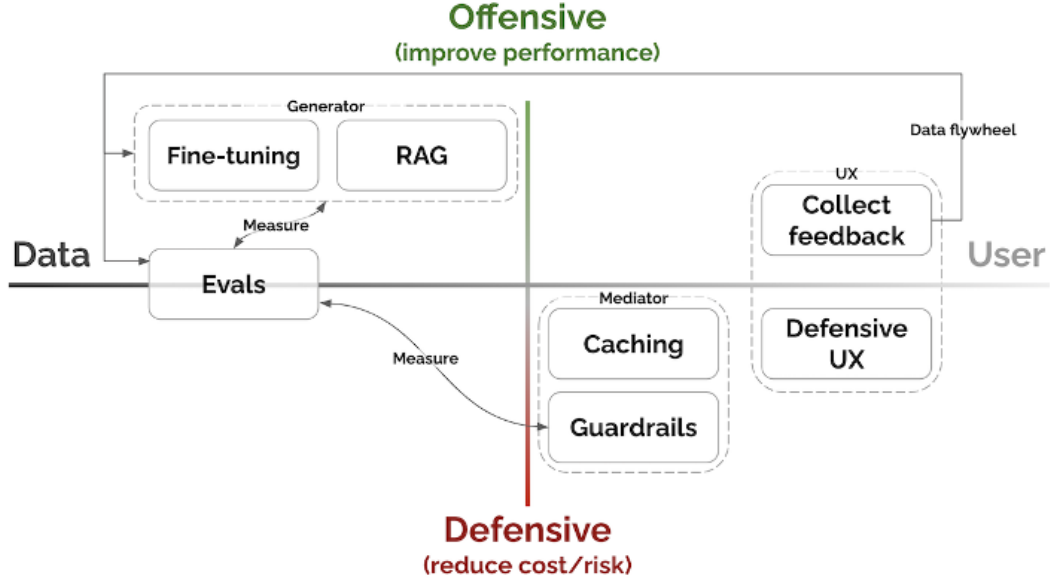


Figure 4: Adaptive Feedback Mechanism in the Learning System

meant easier. This didn't work well. Learners would get three right (sometimes by chance), level up, face harder material, fail, and get frustrated. In early pilot testing, 51% dropped out within three weeks.

We replaced the streak-based approach with confidence-interval tracking. For each learner, we keep a rolling window of the last $N = 20$ attempts. We compute mean performance ($\bar{p}$) and standard error ($SE = \sigma/\sqrt{N}$). Difficulty only increases when the lower bound of the 95% confidence interval exceeds the current level threshold:

$$\text{Advance if: } \bar{p} - 1.96 \times SE > \text{threshold} \quad (1)$$

This ensures that difficulty increases reflect stable improvement, not random variation. The idea is conceptually similar to upper confidence bound algorithms in multi-armed bandit problems (Auer et al., 2002), but applied to educational progression instead of exploration-exploitation trade-offs.

3.5 Experimental Design

Participants: We recruited 120 ESL learners from university language classes ($n = 72$), online language learning forums ($n = 30$), and referrals ($n = 18$). Requirements: age 18-40, non-native English speaker, no pronunciation app use in the past 6 months. Participants received Rs. 500 for completing all assessments. Native languages included Hindi ($n = 50$), Spanish ($n = 22$), Mandarin ($n = 18$), Arabic ($n = 14$), and others ($n = 16$). Mean age was 23.4 years ($SD = 4.2$).

Procedure: The study had three phases. In the pre-test, each participant read 50 sentences

aloud. Recordings were scored by the system and by two human raters (experienced ESL teachers). Inter-rater reliability was high (Cohen's $\kappa = 0.84$). During the six-week training phase, participants used the system at their own pace with a recommendation of 15-20 minutes daily. The system logged all interactions. In the post-test, participants completed the same 50 sentences, again scored by system and human raters.

Ablation conditions: During weeks 3-4, participants experienced three conditions in counter-balanced order: (1) full system, (2) system without adaptive learning (fixed curriculum), and (3) system with template feedback instead of LLM-generated explanations. Each condition lasted one week, with one-week washout between conditions.

Ethics: The study was approved by our institutional review board. Participants gave written informed consent. Audio data is stored on encrypted servers at our institution and will be deleted 12 months after publication.

4. Results

All statistical analyses used Python (scipy 1.9.0, statsmodels 0.13.2) with $\alpha = 0.05$ for significance tests.

4.1 Pronunciation Improvement

Learners improved significantly from pre-test to post-test. Mean accuracy went from 58.3% ($SD = 12.1$) to 76.5% ($SD = 10.8$), a gain of 18.2 percentage points. A paired-samples t-test confirmed this was statistically significant: $t(119) = 8.42$, $p < 0.001$. The effect size was large: Cohen's $d = 0.77$, 95% CI [0.58, 0.96]. This exceeds the average of $d = 0.4$ reported in meta-analyses of computer-assisted language learning (Golonka et al., 2014).

Improvement varied by initial proficiency. Beginners ($n = 65$) improved by $M = 20.1$ percentage points ($SD = 8.3$), while intermediate learners ($n = 55$) improved by $M = 15.8$ percentage points ($SD = 5.9$). Both groups showed significant gains (beginners: $t(64) = 9.21$, $p < 0.001$; intermediate: $t(54) = 6.84$, $p < 0.001$). The difference between groups was also significant ($t(118) = 3.12$, $p = 0.002$), consistent with a ceiling effect for more proficient learners.

4.2 Model Performance

The pronunciation detection model achieved 21% Word Error Rate and 14% Phoneme Error Rate on held-out test data. Performance was consistent across native language backgrounds: WER ranged from 19% (Hindi speakers) to 23% (Arabic speakers). A one-way ANOVA showed no significant differences across language groups ($F(4, 115) = 1.82$, $p = 0.13$).

4.3 Ablation Study

Table 2 shows what happened when we removed each component.

Table 2: Ablation study results ($N = 120$; p-values from paired t-tests)

Condition	Improvement	Retention	p vs Full
Full system	18.2% ($d = 0.77$)	70%	—
Without adaptive learning	14.1% ($d = 0.59$)	58%	< 0.01
Template feedback (no LLM)	12.3% ($d = 0.51$)	65%	< 0.01
Streak-based adaptation (v1)	15.7% ($d = 0.66$)	49%	< 0.001

The confidence-interval adaptation had the biggest impact on retention (70% vs 49% for streak-based, a 21 percentage point difference). LLM feedback contributed most to learning gains compared to templates (18.2% vs 12.3%, a 5.9 percentage point difference). All comparisons with the full system were statistically significant ($p < 0.01$).

4.4 Engagement and Satisfaction

Engagement metrics suggested sustained usage. Mean sessions per week was 4.5 ($SD = 1.8$). Mean session duration increased from 12 minutes in week 1 to 18 minutes in week 6, suggesting growing rather than declining engagement. Overall study completion was 70%, compared to 30-40% typically reported for self-paced language learning apps (Loewen et al., 2019).

User satisfaction was assessed through weekly surveys on a 5-point Likert scale. Mean ratings were: feedback clarity ($M = 4.4$, $SD = 0.6$), perceived improvement ($M = 4.2$, $SD = 0.8$), and system usability ($M = 4.6$, $SD = 0.5$). Qualitative comments frequently mentioned “knowing exactly what to fix” and “fair difficulty progression” as things they liked.

4.5 Phoneme-Level Analysis

Improvement varied by phoneme type. Consonants improved by 22% on average (stops improved fastest at 26%, fricatives slowest at 17%). Vowels improved by 16% (short vowel distinctions like /ɪ/ vs /ɛ/ were hardest). Consonant clusters improved by 18%.

The most difficult phoneme was /θ/ (as in “three”), with only 14% improvement among Hindi and Mandarin speakers whose native languages lack this sound. This aligns with L1 transfer predictions (Flege, 1995) and suggests that sounds absent from L1 may need longer intervention or different teaching approaches.

4.5.1 L1-Specific Error Patterns

We saw distinct error patterns by native language. Hindi speakers ($n = 50$) most often substituted /v/ with /w/ (37% error rate pre-test) and /θ/ with /t_ɳ/ (42% error rate). These patterns reflect Hindi’s phoneme inventory, which lacks /v/ as a distinct sound and has no dental fricatives. After training, /v/ → /w/ errors dropped to 18%, while /θ/ → /t_ɳ/ errors only dropped to 34%, suggesting the dental fricative remains particularly hard.

Spanish speakers ($n = 22$) showed different patterns. The most common error was failing to reduce vowels in unstressed syllables, pronouncing them with full vowels rather than schwa (/ə/). For instance, “about” pronounced as /æbaʊt/ rather than /əbaʊt/. This error dropped from 45% to 28% after training. Spanish speakers also struggled with /b/-/v/ distinctions (31% initial error rate), which improved to 16%.

Mandarin speakers ($n = 18$) showed characteristic errors with final consonants, often deleting them entirely. “Test” would become /tɛs/ or /tɛ/. This makes sense given Mandarin syllable structure, which rarely allows final consonants. Improvement was moderate (22% reduction in final consonant deletion), suggesting this requires more fundamental phonological restructuring that six weeks may not fully achieve.

5. Discussion

5.1 What the Results Mean

The results support our hypothesis that combining phoneme-level detection with explanatory feedback and statistically grounded adaptation produces real learning gains. The effect size ($d = 0.77$) is encouraging and exceeds typical findings in computer-assisted language learning research. But several things are worth discussing.

The ablation study clarifies what each component actually does. Adaptive learning mainly affects retention, not learning speed per session. This makes sense: appropriate difficulty prevents frustration-induced dropout but doesn't directly teach phonetic distinctions. LLM feedback mainly affects learning gains, not retention: explanatory feedback helps learners understand errors but doesn't address motivation.

The confidence-interval adaptation is, I think, a real contribution. The large improvement in retention (70% vs 49%) suggests that statistical grounding matters even though learners don't know about the underlying mechanism. The approach might generalize to other domains where learner performance is noisy and where advancing difficulty too soon causes frustration.

5.2 Limitations

Several limitations should be acknowledged:

Selection bias: People who agreed to a six-week study are probably more motivated than typical app users. Results may not generalize to casual learners who download apps on impulse.

No external control group: We compared pre/post within subjects and ablation conditions, but we didn't include a no-treatment control or comparison with existing commercial apps. Some improvement could be practice effects or just time passing.

Same sentences for testing: Pre and post tests used the same 50 sentences. We didn't directly test transfer to new sentences, though the sentences covered diverse phonetic contexts.

Accent coverage: Model performance was best for Hindi and Spanish speakers and worst for Arabic speakers. This probably reflects imbalances in our training data and suggests we need more diverse non-native speech data.

Audio-only feedback: The system detects errors acoustically but can't tell whether errors come from hearing problems or production problems. Visual feedback showing tongue position would help address this.

5.3 Theoretical Implications

The L1-specific error patterns we observed align with predictions from the Speech Learning Model (Flege, 1995). Phonemes that don't exist in L1 (like /θ/ for Hindi speakers) showed

the smallest improvement, consistent with the idea that establishing new phonetic categories takes more exposure than refining existing ones. This suggests pronunciation training systems should be designed with L1-specific curricula rather than one-size-fits-all approaches.

The success of confidence-interval adaptation also has implications for broader adaptive learning research. Current approaches in intelligent tutoring systems often use binary mastery thresholds or continuous probability estimates. Our findings suggest that explicitly modeling performance variance, rather than treating it as noise to average away, improves learner experience. This might apply to other skill domains with noisy performance metrics, like music practice or athletic training.

5.4 Future Directions

Three directions seem worth pursuing:

Multimodal feedback: Adding visual analysis through lip movement tracking and ultrasound tongue imaging could enable articulatory-level diagnosis. This would distinguish whether errors come from not hearing the difference or not being able to produce it. Each requires different teaching approaches.

Bayesian adaptation: Extending the confidence-interval approach with Bayesian updating could make adaptation more efficient. Instead of requiring a fixed window of 20 observations, a Bayesian model could use prior information about typical learning trajectories to make informed adjustments with less data.

Conversational transfer: Our evaluation used controlled sentence reading. Testing whether gains transfer to spontaneous conversation, where cognitive load is higher, would assess real-world usefulness. This would require developing methods for real-time pronunciation feedback during free-form dialogue.

6. Conclusion

This work addressed the question of how to provide effective, personalized pronunciation training at scale. We built a system combining wav2vec 2.0 phoneme detection, LLM-generated explanatory feedback, and confidence-interval-based adaptive learning. In a controlled study with 120 ESL learners over six weeks, the system produced significant learning gains ($d = 0.77$) and strong retention (70%).

Our main contribution is the confidence-interval adaptation algorithm. Instead of adjusting difficulty based on streak rules, we track statistical stability of performance before advancement. This reduced dropout substantially compared to our initial streak-based approach. The principle might apply beyond pronunciation training to other domains where learner performance is inherently noisy.

On a personal level, this research started because of a client call where my accent seemed to matter more than my technical competence. That bothered me. It still does. But the results here suggest these barriers are addressable. Pronunciation doesn't have to limit opportunities for non-native speakers if training tools can diagnose errors precisely and adapt appropriately to individual learning trajectories.

I want to continue this work in graduate school, particularly exploring multimodal approaches that combine acoustic analysis with visual articulatory feedback. The client call that started this project happened to me, but it happens to millions of non-native speakers every day. If this research can help even some of them, that's worth pursuing.

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